

Git-Vista Security Model

Status: proposed for V2

Git-Vista is local-first and primarily single-user, but it controls real Git repositories and can execute destructive operations. "Only for me" reduces the identity problem; it does not remove browser-origin attacks, hostile repositories, stale tabs, malicious LAN clients, credential leakage, or command races.

Security Objectives

- A website opened in the same browser cannot command the local Git-Vista server.
- A LAN device cannot discover and mutate repositories without explicit pairing.
- The UI cannot request arbitrary paths, Git argv, environment variables, or shell.
- A stale or replayed request cannot repeat a mutation silently.
- Provider and Git credentials never enter logs, URLs, API payloads, or browser storage unnecessarily.
- A repository cannot cause unbounded memory, disk, child-process, or render use. (Implemented for history and file reads: ADR 0022, #63 — every git read goes through `git_vista_git::git_stdout_capped`, which streams under a per-kind cap (8 MiB diff metadata → 413, 200 KB per patch within 5 MB, 2 MB per file) and carries `kill_on_drop`, so a disconnected client kills the child instead of letting it finish into a buffer. History is paged rather than buffered whole, and paging keeps no per-client server state: the entire state of a scroll is one signed offset in the client's cursor, so memory is independent of both repository size and the number of connected clients. The frontend culls to at most 2,000 live rows regardless of camera. The memory bound is a denial-of-service control, not merely a performance measure.)
- Recovery data is protected at least as strongly as the repository itself.
- Security remains understandable for one person to operate.

Assets

- Repository objects, refs, worktrees, index, stashes, and uncommitted changes.
- SSH keys, Git credential-helper material, forge OAuth tokens, and cookies.
- Operation history and recovery refs.
- Repository paths and file contents.
- The ability to push, force-push, create PRs, and modify remote state.

Trust Boundaries

```
flowchart LR
    B[Browser/PWA]
    T[SSH tunnel or HTTPS]
    S[Git-Vista service]
    G[Git subprocess / gix]
    R[(Allowlisted repositories)]
    C[Credential helpers]
    F[Forge APIs]

    B -- untrusted requests --> T
    T --> S
    S -- typed operations only --> G
    G --> R
    G --> C
    S --> F
```

The browser is not trusted merely because it served the UI. Every mutation goes through authentication, origin validation, repository authorization, generation checks, operation planning, and the per-worktree queue.

Operating Modes

Mode	Bind	Transport	Authentication	Intended use
Local	127.0.0.1	HTTP localhost	Launch/session secret + same-origin	Browser on the Linux/ macOS/Windows host
SSH tunnel	127.0.0.1 on Linux	SSH encrypted forwarding	SSH plus Git-Vista session	Primary iPad-to-Linux workflow
LAN view	One explicit interface	Plain HTTP	Single-use bootstrap token, view-scoped read- only routes, rate-limited sign-in	Backup path when the SSH tunnel is unavailable, trusted home LAN only (ADR 0005)
LAN paired, future	Explicit interface	HTTPS	One-time pairing and device session	Trusted private network without SSH tunnel, full read/write
Team, future	Reverse proxy/private network	HTTPS	OIDC/passkeys plus RBAC	Explicit multi-user deployment, not V2 default

Modes are configuration profiles, not a boolean `--public` switch. Each profile has different startup checks and refuses to start when required controls are absent.

Local and SSH Session Design

Implemented by the session + request-protection layer (M1.04, `git-vista-server:: {session, security}`; ADR 0004). The bootstrap token is written `0600` and delivered via the `gv` setup link's URL fragment (never the server, never a log); everything below is enforced by the `require_auth` layer.

- Generate a high-entropy session secret at every service start unless a durable paired-device session is explicitly configured. (256-bit `getrandom` token per start; the in-memory store means a restart is a full revocation. Durable paired sessions deferred to the LAN/paired milestone.)
- Print/open a bootstrap URL whose one-time secret is exchanged for an `HttpOnly`, `SameSite=Strict` session cookie. Remove secrets from the visible URL immediately. (`gv` prints `.../#s=<token>`; the SPA `POST`s it and strips the fragment with `history.replaceState`. The token is single-use — redeeming it rotates a fresh one — and expires.)
- Require a separate CSRF token on every state-changing request. (Per-session token echoed in `x-git-vista-csrf`, compared constant-time; missing/invalid on a live session is a `403`.)
- Validate `Origin`, `Host`, and content type. Reject `Origin: null` mutations. (Host must be a loopback literal — the anti-DNS-rebinding check; `Origin`, when present, must be same-origin and non-`null`; a present content type on a write must be JSON, blocking form-encoded CSRF.)
- Set a strict Content Security Policy and deny framing with `frame-ancestors 'none'`. (See "Browser Security Headers" — stamped on every response by `security_headers`.)
- Bind only to loopback in Local and SSH modes. (Default bind is `127.0.0.1`; the Host/Origin policy is derived from the bind address.)
- Treat localhost as vulnerable to malicious webpages and DNS rebinding; binding loopback is necessary but not sufficient. (The reason the Host allowlist and session gate exist at all.)
- Expire operation approval tokens quickly and bind them to session, repository, worktree, operation hash, and repository generation. (Session/bootstrap expiry landed here; per-operation approval tokens are a later milestone.)

LAN View Profile (implemented, ADR 0005)

`gv --lan-view [path]` starts the existing loopback server plus a second listener, bound to one explicit, operator-confirmed LAN IP:

- The second listener serves a structurally reduced router: GET read routes plus `POST/DELETE /api/session` only. Every write, `/api/select`, `/api/rescan`, `/api/clone`, and `/api/delete-clone` route is never registered on it — absence, not a runtime check (`crates/git-vista-server/ src/main.rs::api_router`).
- `Host/Origin` on this listener are pinned to the one sanctioned LAN IP:port (`security::HostPolicy::lan`); neither `localhost` nor any other address the machine answers on is accepted, so a DNS-rebinding attempt against the LAN listener fails closed the same way the loopback listener's Host check does.

- Sign-in (`POST /api/session`) on this listener is rate-limited per source IP (`crates/git-vista-server/src/ratelimit.rs`); the loopback listener's sign-in is unaffected.
- Auth is otherwise the same single-use bootstrap-token flow as loopback, sharing one in-memory session store; a session established via either listener carries a `via_lan` flag purely so the UI can hide the Active option — the actual write boundary is the LAN router's absent routes.
- Accepted, documented risk: plain HTTP means repo contents and the session cookie are readable by anyone on the same network. Suitable for a trusted home LAN, never a guest or shared network — the startup banner and `gv doctor` say so explicitly.
- `gv doctor` and the launch-time exposed-listener kill-check learn the sanctioned second socket: with `--lan-view`, exactly {loopback, the recorded LAN ip} on port 8080 is healthy; anything else is still a SECURITY ERROR that stops the server. Without the flag, behavior is unchanged from M1.05.

LAN Mode (future, paired HTTPS — write-capable)

The read-only LAN view profile above (ADR 0005) is what's implemented today; this section covers the different, still-future write-capable LAN mode — the "LAN paired" row in the Operating Modes table.

No current write-capable LAN mode exists: the server is hard-limited to `127.0.0.1:8080` plus the read-only LAN view listener above, and the earlier plain-HTTP `--lan` compatibility path was removed. This future mode is a separate paired-HTTPS profile, not a convenience switch, and must:

- Require HTTPS so service workers, credentials, and browser security semantics operate on a secure origin.
- Display a short-lived pairing code locally on the server terminal.
- Pair a browser into a revocable device session; do not use a permanent URL token.
- Show bound interfaces and active paired devices at startup.
- Support revocation and session expiry from the server CLI.
- Rate-limit pairing, authentication, cloning, search, diff, and provider requests.
- Warn that guest Wi-Fi and shared classroom networks are not trusted boundaries.

Future Team Mode

Do not implement Team mode as "LAN mode with more cookies." It requires:

- External identity through OIDC or passkeys/WebAuthn.
- Per-user repository grants and operation attribution.
- Separate workspaces/worktrees for users who may edit concurrently.
- Administrator-controlled repository roots and provider credentials.
- Audit retention, session revocation, quotas, and reverse-proxy deployment.
- A decision about whether Git commits use service identity or user identity.

These concerns are intentionally outside the V2 local-first implementation.

Repository Isolation

Implemented by the server-owned catalog (M1.03, `git-vista-server::catalog`; ADR 0003). The catalog is the only path→id resolver and fails closed on anything it did not itself register.

- Configure one or more allowlisted repository roots. (Catalog `AllowedRoots`; the clones root is always allowed, and a server-launched repo allows its own root.)
- Canonicalize discovered paths and reject escapes through symlinks. (Registration canonicalises the repository root and checks component-wise containment, so a `../` traversal or a symlink escaping an allowed root is rejected.)
- Give the browser opaque repository IDs, not arbitrary filesystem paths. (Requests select a worktree by `WorktreeId`; `GET /api/catalog` reports capabilities by id. A malformed id is a `400`, an unknown id a `404`.)
- Resolve `git-dir` and `common-dir` through Git/gix for normal, bare, and linked worktree repositories. (`git_vista_git::read_repo_facts` classifies each as bare / main / linked and derives the canonical root.)
- Never follow a browser-provided path to open a file or repository. (No endpoint accepts a path; ids resolve only against the registered set.)
- Report capabilities without exposing absolute paths by default. (Descriptors and the graph label carry only a base name unless `GIT_VISTA_EXPOSE_PATHS` is set.)
- Scope operation stores and recovery refs to the canonical repository identity. (Deferred to M1.09.)
- Treat submodules as separate repositories with explicit opt-in traversal. (Deferred.)
- Never serve `.git` internals as static files.

Command Execution

- Use direct argv execution; never invoke a shell. (Enforced by a tripwire test: ADR 0017, #144 — `git-vista-server::argv_boundary` scans both native crates and fails on any process-spawn site that is not allowlisted or does not name `git` literally.)
- Build argv only from typed operation planners and validated domain values. (Extended to read state: ADR 0022, #63 — a paging cursor is server-authored and HMAC-SHA256 signed. The client may echo it back but may not author it. It is validated in a fixed order — length guard, single dot, bounded base64, constant-time tag comparison, only then JSON parse — so a forged or foreign cursor is rejected before `serde_json` sees attacker-shaped bytes and before any repository walk opens. A cursor scoped to a different repository or worktree returns the generic `400`, deliberately not a distinguishing error, so probing cannot confirm that another target exists.) (Implemented for every served-repository mutation: ADR 0015/0016, #142/#143 — write handlers build a typed `GitOperation`, and `git-vista-server::planner` is the only place a mutating git argv is constructed. Proven at the API boundary by adversarial fixtures: ADR 0017, #144 — no route deserializes a raw command string or argv array, and hostile bodies die at the extractor.)
- Pass `--` where Git supports it and validate full refnames with Git.

- Clear or explicitly set child environment variables. Disable terminal prompts, editors, pagers, and hooks where the operation semantics permit.
- Decide hook policy explicitly. Running repository hooks may execute arbitrary local code; local mode may allow them, but the UI must report that fact and Team mode should default to a restricted policy.
- Apply timeouts, cancellation, stdout/stderr limits, process-group termination, and concurrency quotas.
- Convert raw Git errors into structured safe errors; retain detailed stderr only in local protected logs with credential redaction.

Remote and Forge Credentials

- Prefer existing Git credential helpers and SSH agents on the Linux host.
- Never ask the browser to upload an SSH private key.
- Store forge tokens server-side using the OS keyring or an encrypted local store.
- Prefer fine-grained, short-lived provider credentials. GitHub recommends GitHub Apps over OAuth apps for tighter permissions and short-lived tokens.
- Request provider scopes per capability and show them before authorization.
- Redact URL userinfo, HTTP authorization, query tokens, and credential-helper output from logs and operation records.
- Make provider logout revoke/delete local credentials.

See [GitHub's app authentication guidance](#).

Request Integrity

Every mutation request contains:

- Session and CSRF proof.
- Repository and worktree IDs.
- Expected repository generation.
- An idempotency key generated by the client.
- A typed operation body.
- For risky actions, a short-lived approval token from a prior plan.

The server records idempotency results for a bounded period. A retried mobile request returns the original result rather than performing the action twice.

(Generation, hash and expiry are enforced at execution time: ADR 0018, #145 — `planner::validate` refuses a tampered operation hash and an expired plan, and `planner::enforce_fresh` recomputes the repository generation (HEAD, all refs, worktree status) and re-verifies every build-time-held precondition against the live repository immediately before execution. Any drift refuses with a 409 and a client-facing reason — the TOCTOU gap fails closed. Client-supplied generation/idempotency fields arrive with the review roundtrip, M2+.)

Operation Risk Classes

Class	Examples	Required control
Read	History, diff, blame	Authentication, limits, path isolation
Local reversible	Stage, branch create	Serialized operation, generation check
History rewrite	Reset, rebase, amend	Preview, explicit confirmation, recovery ref
Worktree destructive	Clean, discard, checkout overwrite	Preview, typed file impact, safety checkpoint
Remote visible	Push, PR create/comment	Explicit target and identity confirmation
Remote destructive	Force-push, remote branch delete	Strong warning, lease/CAS, re-auth option

No gesture, pressure threshold, swipe, or double-tap directly executes a destructive operation. Touch gestures may select or open a plan; final confirmation is explicit.

Clone and Network Controls

- Restrict clone schemes by operating mode and configuration.
- Block loopback, link-local, cloud metadata, and private-network destinations when arbitrary URL cloning is exposed beyond local mode, unless explicitly allowlisted.
- Apply clone size/time quotas and stream progress without buffering complete output.
- Store temporary clones under a managed root with ownership metadata and expiry. (Implemented: ADR 0008, #121 — clones persist under `$XDG_DATA_HOME/git-vista/clones`, not a wiped-at-startup temp dir; deletion is explicit via `POST /api/delete-clone`, guarded to paths that canonicalize inside that root.)
- Never delete a clone while an active repository handle references it. (Implemented: ADR 0008, #121 — `delete_clone` refuses the currently open selection with `409`.)
- Treat remote URLs as secrets because they can contain credentials.

Browser Security Headers

At minimum:

```
Content-Security-Policy: default-src 'self'; object-src 'none'; base-uri 'none';  
    frame-ancestors 'none'; form-action 'self'; connect-src 'self'  
Cross-Origin-Opener-Policy: same-origin  
Cross-Origin-Resource-Policy: same-origin  
Referrer-Policy: no-referrer  
X-Content-Type-Options: nosniff  
Permissions-Policy: camera=(), microphone=(), geolocation=()  
Cache-Control: no-store # authenticated API responses
```

Static fingerprinted assets may be immutable. Do not apply `no-store` blindly to the PWA shell and destroy offline startup.

Implemented by `security_headers` (M1.04; ADR 0004), stamped on every response. Two deviations from the baseline above, both documented in ADR 0004: the CSP's `script-src` adds `'wasm-unsafe-eval'` (required by the WebAssembly runtime) and `'unsafe-inline'` (Trunk boots the wasm with an inline module script whose hash changes each build, and the server sets a static header), and `img-src/font-src` are widened to `'self'` `data: /` `'self'` for the inline SVG favicon and the bundled Nerd Font. `Cache-Control: no-store` is applied to API responses only; the SPA shell keeps `no-cache` so offline startup (a later PWA milestone) survives.

Data at Rest

- Operation history may expose commit messages, branches, and repository paths.
- Default to user-only filesystem permissions on the Linux host.
- Keep browser persistence minimal and clearable.
- Do not cache private diffs/file contents offline by default.
- Recovery refs and safety stashes require retention limits and visible cleanup.
- Provider tokens and device sessions must not be stored in the Git repository.

Audit and Logging

Log structured events with operation ID, repository ID, worktree ID, risk class, duration, result code, and before/after generation. Do not log request bodies by default. Local logs should be useful without exposing credentials or complete file contents.

Security Testing

- Route tests for missing/invalid session, CSRF, Origin, Host, content type, and repository access.
- Concurrency tests for clone switching, stale generations, duplicate requests, rebase/reset interleaving, and two browser tabs.
- Property/fuzz tests for ref, path, status, diff, and provider parsers.
- Tests for symlink escapes, linked worktrees, bare repositories, submodules, and repositories with hostile names/messages.
- Output-limit and timeout tests using intentionally noisy/slow child processes.
- Browser tests for cookie and service-worker update behavior.
- Dependency, license, secret, and vulnerability scanning in release CI.

Known Non-Goals

- Protecting a repository from its own Unix account owner.
- Sandboxing arbitrary Git hooks in Local mode.
- Providing tenant isolation in V2.
- Making remote force-push universally undoable.
- Securing plain HTTP LAN mode; it should not exist as a supported write mode.

Security Release Gate

No release should claim safe remote or LAN write access until:

- Loopback is the default bind.
- SSH tunnel mode is documented and tested on iPad.
- Every mutation has session, CSRF, origin, generation, and serialization checks.
- Repository paths are allowlisted and opaque to the browser.
- Child processes have time/output limits and credential redaction.
- Recovery behavior is documented for each destructive operation.
- A `SECURITY.md` vulnerability-reporting policy exists.